

EE 372 Random Signals

Assignment #5

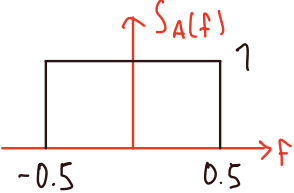
Due date/time: May 11, 2025 23:59

Suppose that $X(t)$ is a zero-mean Gaussian random process with autocorrelation function $R_X(\tau) = \text{sinc}^2(\tau)$. Suppose that I run $X(t)$ through an ideal lowpass filter with unity gain and bandwidth 0.5 Hz (that is, $H(f) = 1$ for $|f| < 0.5$ and zero otherwise) and obtain $Y(t)$ at the output.

- As a mathematical helper function, let us define $S_A(f) = 1$ for $|f| < 0.5$ and zero otherwise. Calculate its inverse Fourier transform by hand and obtain the corresponding time-domain function. Express (or draw) the power spectral density $S_X(f)$ utilizing this function.
- Calculate $P[Y(0) \geq 2]$.

* Firstly, note that $X(t)$ is a WSS random process since $\mu_X = 0$, $R_X(\tau)$ are independent of t .
 $\hookrightarrow$ Thus, $Y(t)$ is WSS as well

a)



$$S_A(f) = \begin{cases} 1, & |f| < 0.5 \\ 0, & \text{o.w} \end{cases}$$

$$S_A(t) = \mathcal{F}^{-1}\{S_A(f)\} = \int_{-0.5}^{0.5} e^{j2\pi ft} df = \frac{1}{j2\pi t} \left[e^{j2\pi ft} \right]_{-0.5}^{0.5}$$

$$= \frac{1}{\pi t} \cdot \frac{e^{j\pi t} - e^{-j\pi t}}{2j} = \frac{\sin(\pi t)}{\pi t} = \boxed{\text{sinc}(t)}$$

By Wiener - khintchine theorem we know that

$$S_X(f) = \int_{-\infty}^{\infty} R_X(\tau) e^{-j2\pi f\tau} d\tau = \mathcal{F}\{R_X(\tau)\} = \mathcal{F}\{\text{sinc}^2(\tau)\}$$

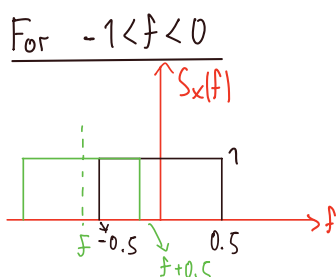
$\rightarrow$ Recall that Fourier of multiplication of 2 time domain signals equals to their convolution in freq. domain.

$$\mathcal{F}\{X(t)H(t)\} = X(f) * H(f) = \int_{-\infty}^{\infty} X(f') H(f-f') df'$$

* Then $\mathcal{F}\{\text{sinc}^2(\tau)\} = \mathcal{F}\{\text{sinc}(\tau)\} * \mathcal{F}\{\text{sinc}(\tau)\}$

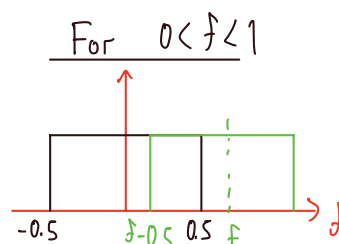
$\hookrightarrow$ BUT we know that $\mathcal{F}^{-1}\{S_A(f)\} = \text{sinc}(\tau)$ or $\mathcal{F}\{\text{sinc}(\tau)\} = S_A(f)$

Then $\boxed{S_X(f) = S_A(f) * S_A(f)} = \int_{-\infty}^{\infty} S_A(f') S_A(f-f') df'$



$$S_X(f) = \int_{-0.5}^{f+0.5} 1 df$$

$$= f + 1$$

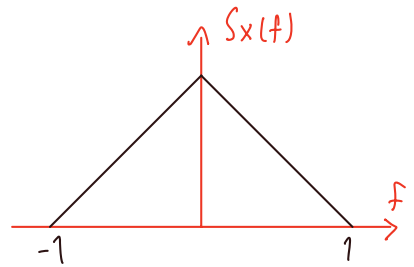


$$S_X(f) = \int_{f-0.5}^{0.5} 1 df$$

$$= 1 - f$$

Also notice that $S_X(f) * S_X(f) = 0$ for $f < -0.5$ and $f > 0.5$

$$S_X(f) = \begin{cases} 1-f, & 0 < f < 1 \\ 1+f, & -1 < f < 0 \\ 0, & \text{o.w.} \end{cases}$$

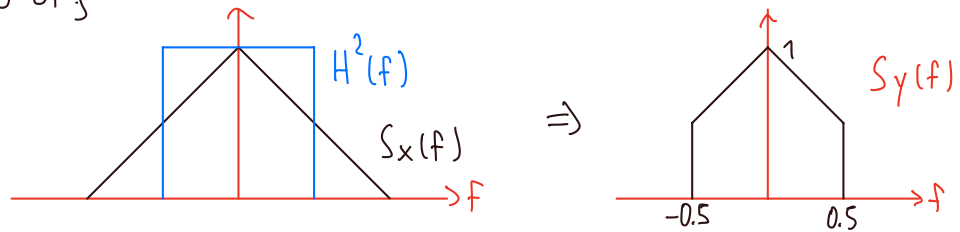


b) We have remarked that $Y(t)$ is a Gaussian WSS random process as well
 ↳ Also, from Theorem 10.15, $Y(t)$ is a stationary random process

$$P[Y(0) > 2] = Q\left(\frac{2 - \mu_Y}{\sigma_Y}\right) = Q\left(\frac{2}{\sigma_Y}\right) \Rightarrow \text{Since } Y(t) \text{ is a Gaussian random process } Y(0) \text{ is a Gaussian random variable}$$

$$\text{Note that } \text{Var}[Y(0)] = \sigma_Y^2 = E[Y(0)^2] = R_Y(0) = \int_{-\infty}^{\infty} S_Y(f) df$$

Passing $X(t)$ through LTI filter $H(f)$ yields a PSD of $\left. \begin{array}{l} S_Y(f) = |H(f)|^2 S_X(f) = \text{rect}_{0.5}^2(f) * \text{tri}_1(f) \end{array} \right\}$



$$S_Y(f) = \begin{cases} 1-|f|, & |f| < 0.5 \\ 0, & \text{o.w.} \end{cases}, \quad R_Y(0) = \int_{-0.5}^{0.5} (1-|f|) df = 2 \int_0^{0.5} (1-f) df$$

$$= 2 \left[f - \frac{f^2}{2} \right]_0^{0.5} = 2 \left(0.5 - \frac{0.25}{2} \right) = 0.75$$

$$\text{Then, } \sigma_Y^2 = \frac{3}{4} \Rightarrow \sigma_Y = \frac{\sqrt{3}}{2}$$

$$P[Y(0) > 2] = Q\left(\frac{4}{\sqrt{3}}\right) \approx 0.0105$$